

26TH ANNIVERSARY OF
PCI-IOSS
SPINE SURGERY SUMMIT
in conjunction with
ISASS AP 2024

INTERNATIONAL SOCIETY FOR THE ADVANCEMENT OF SPINE SURGERY ASIA-PACIFIC SECTION

28 - 31 AUGUST 2024 | YOGYAKARTA, INDONESIA

Incidence and risk factors for mechanical failure
after ACR in adult spinal deformity surgery

Se-Jun Park
Samsung Medical Center
Seoul, Korea



WWW.PCI-IOSS-ISASSAP2024.COM

ACR : High vulnerability to mechanical failure



Cut ALL
→ Loss of anterior tension band



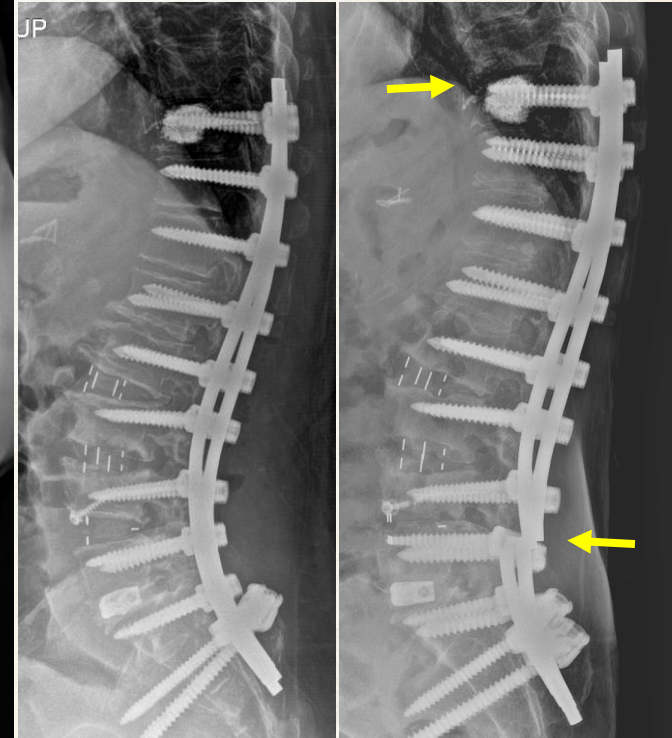
Pseudoarthrosis & rod fracture

Preoperative severe deformity
→ Large amount of correction



PJK / PJF

To investigate the incidence and risk factors of mechanical failure (MF) after ACR in ASD surgery



Study design

- Inclusion criteria
 - Patients with DSI : PI-LL $\geq 20^\circ$
 - ≥ 1 -level ACR, ≥ 5 -level fusion to pelvis; ≥ 2 -year FU
- Mechanical failure
 - PJK : PJA $> 28^\circ$, Δ PJA $> 22^\circ$
 - Rod fracture : only at ACR levels
 - Revision surgery for PJK or rod fracture

Results

- 147 patients
 - 70.3 years; 126 females (90.6%)
 - ACR for 236 levels (mean 1.6 levels per patients)
 - Median fusion length : 8 levels
- Mechanical failure rates
 - *MF : 49 patients (33.3%) → revision for 28 patients (19.0%)*
 - PJK : 41 patients(27.9%) → revision for 19 patients
 - RF : 11 patients (7.5%) → revision for 10 patients

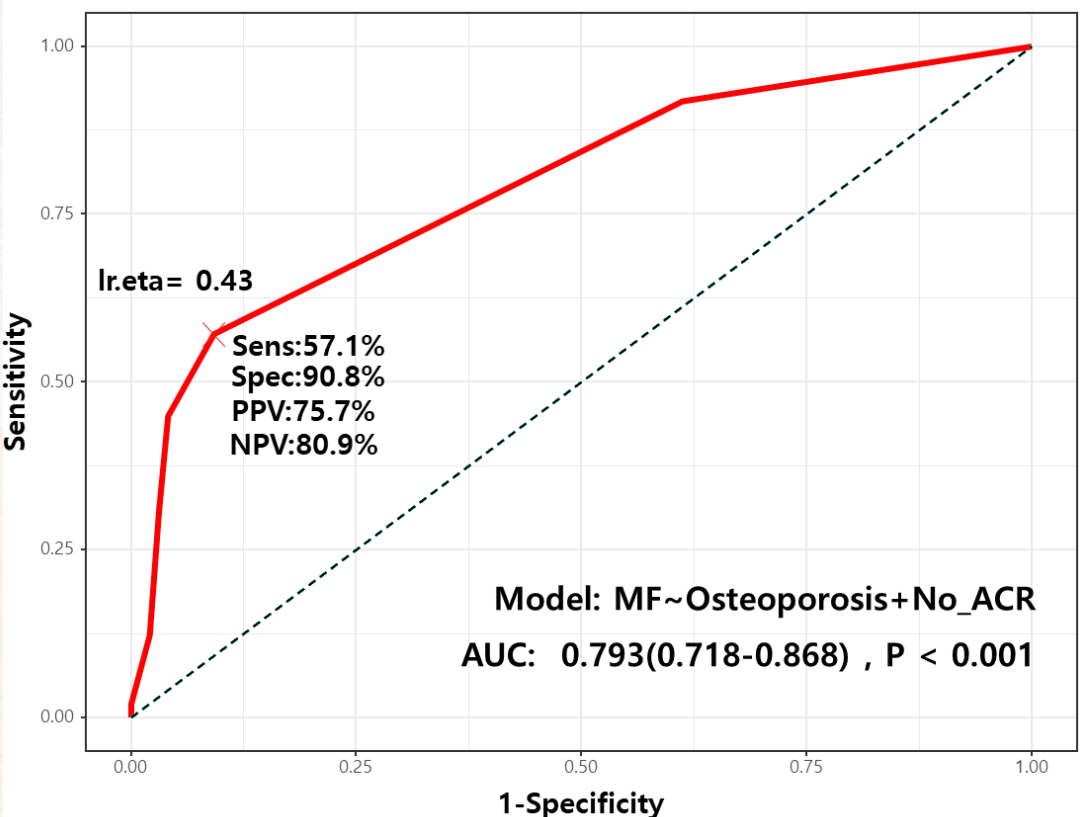
YOGYAKARTA - INDONESIA	No-MF (N=98)	PJK (N=41)	RF (N=11)	P1*	P2*
<i>Demographic variables</i>					
Sex: female, n (%)	88 (89.8%)	38 (92.7%)	10 (90.9%)	0.694	0.842
ASA score	2.08 ± 0.42	2.15 ± 0.42	1.91 ± 0.30	0.520	0.238
BMI (kg/m ²)	26.3 ± 4.7	26.6 ± 3.2	25.6 ± 3.8	0.609	0.635
T-score	-1.5 ± 1.1	-1.7 ± 1.1	-1.2 ± 1.0	0.340	0.502
Osteoporosis, n (%)	13 (13.3%)	13 (31.7%)	1 (9.1%)	0.017	0.235
Preoperative teriparatide, n (%)	21 (21.4%)	13 (31.7%)	0 (0%)	0.269	0.080
Smoking, n (%)	3 (3.1%)	1 (2.4%)	1 (9.1%)	0.850	0.527
DM, n (%)	20 (20.4%)	6 (14.6%)	2 (18.2%)	0.376	0.122
<i>Operative variables</i>					
Revision case, n (%)	35 (35.7%)	12 (29.3%)	4 (36.4%)	0.535	0.961
ALIF at L5-S1, n (%)	53 (54.1%)	19 (46.3%)	5 (45.5%)	0.397	0.645
Number of ACR levels	1.7 ± 0.6	1.9 ± 0.8	2.2 ± 0.6	0.105	0.019
Cage subsidence, n (%)	6 (6.1%)	3 (7.3%)	1 (9.1%)	0.794	0.703
Number of interbody fusion	3.8 ± 0.8	3.9 ± 0.7	3.6 ± 0.7	0.959	0.288
Total levels fused	6.8 ± 1.9	6.1 ± 1.7	8.7 ± 1.8	0.019	0.001
Cement at UIV, n (%)	43 (43.9%)	19 (46.3%)	6 (54.5%)	0.946	0.386
TP hook at UIV+1, n (%)	19 (19.4%)	4 (9.8%)	3 (27.3%)	0.061	0.393
Satellite rods, n (%)	22 (22.4%)	7 (17.1%)	0 (0%)	0.424	0.079

	No-MF (N=98)	PJK (N=41)	RF (N=11)	P1*	P2*
Preoperatively					
PI (°)	56.1 ± 10.7	55.1 ± 9.6	52.1 ± 10.9	0.523	0.278
LL (°)	6.2 ± 18.7	7.0 ± 17.5	6.4 ± 14.6	0.892	0.979
PI-LL (°)	49.9 ± 17.4	48.1 ± 15.8	45.7 ± 15.5	0.625	0.738
SS (°)	19.6 ± 11.5	19.1 ± 10.9	14.2 ± 7.6	0.810	0.234
PT (°)	36.5 ± 10.4	35.9 ± 9.2	37.9 ± 10.9	0.680	0.742
TK (°)	8.3 ± 12.3	8.9 ± 12.6	9.9 ± 15.9	0.908	0.832
TPA (°)	37.7 ± 11.5	36.9 ± 8.9	39.7 ± 10.9	0.610	0.786
C7–SVA (mm)	96.0 ± 59.4	102.5 ± 58.4	95.1 ± 57.3	0.582	0.988
C7–CSVL (mm)	17.2 ± 14.7	17.9 ± 14.1	27.5 ± 19.7	0.721	0.069
Postoperatively					
LL (°)	51.2 ± 10.4	50.3 ± 10.4	53.3 ± 13.5	0.635	0.441
LDI (%)	57.6 ± 19.1	61.4 ± 18.2	53.7 ± 18.9	0.366	0.600
PI-LL (°)	4.8 ± 9.7	4.4 ± 9.0	-0.7 ± 15.9	0.808	0.090
SS (°)	38.4 ± 8.3	34.9 ± 8.7	38.9 ± 8.4	0.022	0.741
PT (°)	17.5 ± 8.9	20.0 ± 6.5	12.9 ± 13.2	0.119	0.115
TK (°)	24.8 ± 10.6	29.9 ± 11.3	26.5 ± 10.9	0.019	0.537
TPA (°)	14.3 ± 8.5	17.1 ± 7.5	10.1 + 11.6	0.082	0.133
C7–SVA (mm)	17.4 ± 31.3	17.3 ± 29.6	18.0 ± 37.3	0.955	0.988
C7–CSVL (mm)	13.9 ± 11.8	11.0 ± 8.7	24.4 ± 15.0	0.171	0.007

	P1 (PJK)	P2 (RF)
<i>SRS-Schwab's classification</i>		
PI-LL modifier	0.828	0.910
PT modifier	0.830	0.837
SVA modifier	0.924	0.642
<i>GAP scoring system</i>		
Total GAP score (points)	0.398	0.291
GAP score category	0.873	0.425
<i>Age-adjusted alignment†</i>		
Age-adjusted PI-LL	0.394	0.392
Age-adjusted PT	0.274	0.871
Age-adjusted TPA	0.165	0.966
<i>Roussouly, restored</i>	0.252	0.134

Multivariate regression model for MF

Variables	P	Odds ratio	95% CI
Osteoporosis	0.048	2.361	1.270 – 5.590
Number of ACR level	0.036	1.762	1.039 – 3.587



Subgroup analysis of rod fracture

	RF (-) (N=250)	RF (+) (N=13)	P
<i>Operated levels</i>			0.340
L1-2, n (%)	3 (1.2%)	0 (0%)	
L2-3, n (%)	62 (24.8%)	6 (46.2%)	
L3-4, n (%)	103 (41.2%)	3 (23.1%)	
L4-5, n (%)	82 (32.8%)	4 (30.8%)	
<i>Osteotomy grade</i>			0.912
Grade 1, n (%)	116 (46.4%)	6 (46.2%)	
Grade 2, n (%)	85 (34.0%)	5 (38.5%)	
Grade 2 + laminectomy, n (%)	49 (19.6%)	2 (15.4%)	

Conclusion

Proximal junctional kyphosis

- Osteoporosis
- UIV < T10
- Greater postoperative TK
- No use of TP hook (ns)

Rod fracture

- Number of ACR levels
- Total fusion levels
- No use of teriparatide (ns)
- No use of satellite rods (ns)

Conclusion

Proximal junctional kyphosis

- Osteoporosis
- UIV < T10
- Greater postoperative TK
- No use of TP hook (ns)

Rod fracture

- Number of ACR levels
- Total fusion levels
- No use of teriparatide (ns)
- No use of satellite rods (ns)

UIV $\geq$ T10 with TP hook
Use of perioperative TPH
Use of satellite rod

➤ Spine (Phila Pa 1976). 2024 Aug 1. doi: 10.1097/BRS.0000000000005114. Online ahead of print.

Incidence and Risk Factors for Mechanical Failure after Anterior Column Realignment in Adult Spinal Deformity Surgery

Se-Jun Park¹, Jin-Sung Park¹, Minwook Kang¹, Kyunghun Jung¹, Chong-Suh Lee²,
Dong-Ho Kang¹

Thank you for your kind attention!!